PAUL JUSTIN CONNOR SMITH

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SUMMARY

Cognitive neuroscientist with 3+ years of experience analyzing large, high-dimensional electrophysiological datasets. Builds reproducible analysis pipelines and applies machine learning and multivariate statistics in Python, R, and MATLAB.

EXPERIENCE

University of Münster — 2022 - Present
Research Associate, Experimental Psychology — *Münster*

- Doctoral research on EEG oscillations and visual memory: design experiments and build end-to-end analysis pipelines for large multichannel EEG and eye-tracking datasets in MATLAB, Python, and R.
- Apply multivariate methods — decoding/MVPA, mixed-effects and linear models, time–frequency analysis — to high-dimensional neural data.
- Develop reproducible, version-controlled (Git) analysis workflows for collaborative research.
- Teach research methods, neuroscience methods, and decision-making to B.Sc. Psychology and M.Sc. Neuroscience students.

University of Münster — 2021 - 2022
Research Assistant, Experimental Psychology — *Münster*

- Collected and preprocessed EEG data and contributed to the EEG ManyPipelines project — a multi-laboratory study quantifying how analytic pipeline choices affect scientific conclusions (analytic robustness and reproducibility).

University of Münster — 2019 - 2021
Research Assistant, Organisational & Business Psychology — *Münster*

- Built and applied machine-learning models in R and Python to evaluate the use of ML in business-psychology research.
- Ran statistical analyses on behavioral and survey data and helped implement reusable analysis code.

German Bundestag (19th) — 2019
Intern, Office of MdB Kordula Schulz-Asche — *Berlin*

- Supported healthcare-policy work and research on modernization of the healthcare sector; synthesized technical material for non-specialist stakeholders.

University Hospital Münster — 2017 - 2019
Research Assistant, Biomagnetism & Biosignalanalysis — *Münster*

- Conducted MEG recordings and analyzed neural data using linear models in R and MATLAB for research on tDCS therapy and fear generalization.

Precire Technologies GmbH — 2019
Intern — *Aachen*

- Applied machine learning in Python to analyze natural-language patterns in an industry NLP setting.

TECHNICAL SKILLS

Programming & tools	Python, R, MATLAB, SPSS, Git
Machine learning & stats	Classification & regression, cross-validation, dimensionality reduction, multivariate decoding (MVPA), mixed-effects & linear models, scikit-learn
Data & signal processing	Time–frequency analysis, spectral methods, filtering, large multichannel dataset wrangling (pandas, NumPy), eye-tracking

PUBLICATIONS

Smith, P.J.C., & Busch, N.A. (2025). *Spontaneous alpha-band lateralization extends persistence of visual information in iconic memory by modulating cortical excitability.* Journal of Neuroscience.

Smith, P.J.C., & Busch, N.A. (2026). *Effects of spatial attention on iconic memory are primarily driven by costs rather than benefits.* bioRxiv (preprint).

Smith, P.J.C., & Busch, N.A. (2025). *Pre-stimulus pupil-linked arousal enhances initial stimulus availability and accelerates decay in iconic memory.* bioRxiv (preprint).

McCain, K.J., Friedl, W.M., Pourtois, G., Busch, N., **Smith, P.J.C.**, Van Migem, M., Wiens, S., Adamian, N., Mushtaq, F., Pavlov, Y.G., & Keil, A. (2025). *Effects of spatial attention on visuocortical processing: a multi-laboratory replication of Clark & Hillyard (1996).* PsyArXiv (preprint).

EDUCATION

University of Münster 2022 - Present
PhD in Cognitive Neuroscience, advisor Prof. Dr. Niko Busch.
Focus: interaction of the alpha rhythm of the human EEG and iconic memory performance.

University of Münster 2019 - 2022
M.Sc. in Cognitive Neuroscience (1.2)
Thesis: "Temporal Dynamics of Decision Making and Confidence." (1.0)

University of Münster 2016 - 2019
B.Sc. in Psychology (1.7)
Visiting student, **Columbia University**, New York — DAAD PROMOS Scholarship (2018)
Thesis: "Predicting Conditioned Stimuli in Spider Phobics: Machine Learning Applications on Electrophysiological Data." (1.3)

Graf Adolf Gymnasium Tecklenburg 2008 - 2016
Abitur: 1.0